3 A trolley of mass m is held on a smooth horizontal surface by means of two identical springs, each of force constant k and unstretched length L. The springs are attached to fixed points as shown in Fig. 3.1.

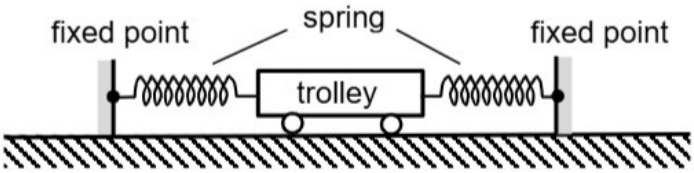


Fig. 3.1

(a) When the trolley is in equilibrium, the extension of each spring is e.

(i) The trolley is displaced horizontally to the right and then released at time $t = 0$. The trolley then oscillates with simple harmonic motion. The springs remain stretched throughout the oscillation.

Show that the acceleration a of the trolley is given by

$$a = -\frac{2k}{m}x$$

where x is the displacement of the trolley from its equilibrium position.

Explain your working.

[2]

(ii) The mass of the trolley is 0.63 kg and the force constant of each spring is 50 N m^{-1}.

1. Determine the period T of the oscillations.

T = ... s [2]

2. Determine the maximum speed of the trolley if the trolley is displaced 4.0 cm from its equilibrium position before released.

maximum speed = ... $m\ s^{-1}$ [1]

3. On Fig 3.2, sketch a graph to show the variation with t of the kinetic energy E_k of the trolley for one complete oscillation, starting at $t = 0$.

Include all relevant values.



Fig 3.2

[2]

(b) One additional identical spring is added on each side of the trolley as shown in Fig 3.3. The trolley is then made to oscillate.

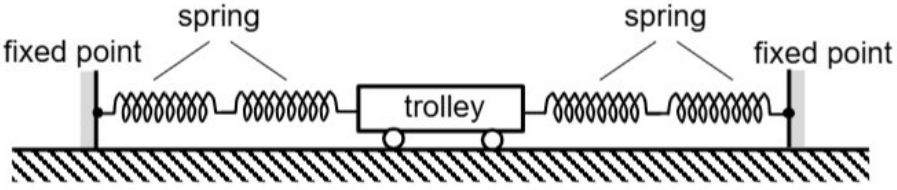


Fig 3.3

State and explain the change, if any, in the period of oscillation of the trolley.

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..... [2]

[Total: 9]